

PETALNEX PVT LTD

Internship — Daily Progress Report

Intern Name	Muhammad Bin Waseem
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Daily Report Fields

Today I Learned	I learned why AI automation workflows require structured and predictable outputs instead of free-form text. I learned how to use a Structured Output Parser with a JSON schema in n8n, how to define required fields and allowed values, and how validation and fallback handling can prevent unexpected AI responses from breaking a workflow.
Today I Built / Practiced	I upgraded my AI Email Classification workflow from Day 15 to use structured AI output. The workflow now generates a JSON object containing category, priority, sentiment, department, and summary , validates the returned values, and routes valid results to the appropriate category. I also added a fallback branch for invalid responses and successfully tested the fallback by temporarily changing the accepted Spam category to an invalid value, causing the validation to fail and the fallback branch to execute.
Technologies / Tools Used	n8n, Groq Chat Model, Basic LLM Chain, Structured Output Parser, JSON Schema, IF/Validation Logic, Switch Node, Edit Fields, Prompt Engineering
Problem(s) Faced	The main challenge was verifying that the fallback branch could actually be triggered, since the Structured Output Parser normally ensures that the model returns data matching the defined schema.
How I Solved Them	I temporarily changed the accepted Spam value in the validation rule to an invalid value. When the AI returned Spam, the validation condition failed, which successfully routed the workflow to the fallback branch. After confirming the fallback

	worked, the validation rule could be restored to its original configuration.
Task Status	Completed
GitHub / Workflow Link	https://github.com/MuhammadBinWaseemm/AI-Automation-Internship-Pentalnex-Pvt.-Ltd/tree/main/daily-logs/DAY-16
Plan for Next Day	Looking forward to Day-17.